



Applications such as video on the Internet require all the packets of data that it is made up of to arrive at the same time. When this does not happen you end up with output that is of unacceptable quality. Because the network is unable to distinguish between different kinds of data, all packets of information are treated similarly. So a packet containing e-mail data is given the same priority as more demanding applications such as high-resolution video. Whenever network congestion takes place, these data packets are eliminated/lost indiscriminately. Also, there's never really a guaranteed connection with the Internet. QoS technology, when implemented in routers, will help address some of these concerns because it allows prioritisation of packets to advanced applications and will allow them to be pushed through the network faster than, say, a simple e-mail. This 'customisation' of information flow will lead to a more reliable network. In time this will mean that ISPs will offer you connections with a speed and QoS guarantee, unlike today when you just have to live with what you get.

As our dependence on technology continues to increase, a large number of devices that make life simpler will require to be connected to the network all the time. These devices, along with the increasing number of computers that make up the Internet, will require a huge number of constant (or static) IP addresses. IPv6 (www.ipv6.org) is an upgrade to the existing Internet Protocol (IPv4) that will overcome the limitations that the small address space and other limitations of IPv4 cause. It uses a 128-bit address field instead of the 32-bit field used by IPv4, which would make enough unique addresses available for the countless intelligent devices that are expected to log onto networks over the next few years. Other changes include

improvements that make it possible for new machines entering a network for the first time to configure themselves automatically. IPv6 is also designed to improve upon IPv4's security (with IPSec), reliability and network management.

Gateways to other networks

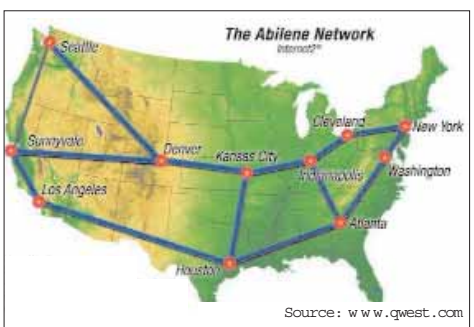
Transferring voice over data networks has become a preference for both corporate and public network planners. This allows considerable

Internet2: A Case Study in Progress

The Internet2 project is a consortium consisting of 180 universities in the US, that are working in partnership with both industry and the government to deploy advanced network technologies and develop applications that will run on them. Members of the Internet2 project are connected either directly or via GigaPoPs, which are high-speed access points located in different parts of the US, to the very high performance Backbone Network Service (vBNS) and the Abilene network. The latter is another high-performance network developed and operated by Internet2, Cisco Systems, Nortel Networks, Qwest Communications and Indiana University.

The Abilene network is operating initially on 2.4 Gbps backbone links but researchers are working on pushing this to 9.6 Gbps and beyond. The minimum

connection speed for users of Internet2 is 155 Mbps, which is about 3,000 times faster than dial-up Internet connections. Applications such as broadcast-quality video streaming and distributed instruction scenarios have been tested for the first time on this



closed network. Abilene and a few other networks in different parts of the world are the only places that you can get a real-life peek into the Internet's future.

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